

M-Estimators

1.1 Definition

Definition 1.1. For any function $\psi(\mathbf{y}; \boldsymbol{\theta})$, define the *functional* $T(F) = \boldsymbol{\theta}_0$, where $\boldsymbol{\theta}_0$ is a solution to the equation:

$$\Psi_F(\boldsymbol{\theta}_0) \equiv \int \psi(\mathbf{y}; \boldsymbol{\theta}_0) dF(\mathbf{y}) = 0. \quad (1.1)$$

Suppose $(\mathbf{y}_i)_{i=1}^n$ is a random sample from distribution F , and let $\mathbb{F}_n$ denote the empirical distribution function. The **M-estimator** corresponding to ψ is $\hat{\boldsymbol{\theta}}_n = T(\mathbb{F}_n)$, which is a solution to the equation:

$$\Psi_n(\hat{\boldsymbol{\theta}}_n) = \int \psi(\mathbf{y}; \hat{\boldsymbol{\theta}}_n) d\mathbb{F}_n(\mathbf{y}) = \frac{1}{n} \sum_{i=1}^n \psi(\mathbf{y}_i; \hat{\boldsymbol{\theta}}_n) \stackrel{\text{Set}}{=} \mathbf{0}.$$

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Discussion 1.1. Often, $\psi(\mathbf{y}; \boldsymbol{\theta})$ arises as the gradient of an objective function $q(\mathbf{y}; \boldsymbol{\theta})$:

$$\psi(\mathbf{y}; \boldsymbol{\theta}_0) = \frac{\partial}{\partial \boldsymbol{\theta}_0} q(\mathbf{y}; \boldsymbol{\theta}_0),$$

and $\boldsymbol{\theta}_0$ solving (1.1) is a solution to the minimization problem:

$$\boldsymbol{\theta}_0 = \arg \min_{\boldsymbol{\theta} \in \Theta} \int q(\mathbf{y}; \boldsymbol{\theta}) dF(\mathbf{y}).$$

For example, suppose F is a parametric distribution $F(\mathbf{y}; \boldsymbol{\theta}_0)$ and the goal is to estimate $\boldsymbol{\theta}_0$. Consider specifying the negative log likelihood $-\ln f(\mathbf{y}; \boldsymbol{\theta})$ as the objective $q(\mathbf{y}; \boldsymbol{\theta})$. The gradient $\psi(\mathbf{y}; \boldsymbol{\theta})$ of $q(\mathbf{y}; \boldsymbol{\theta})$ is the parametric score equation:

$$\Psi_n(\boldsymbol{\theta}) = -\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \boldsymbol{\theta}} \ln f(\mathbf{y}_i; \boldsymbol{\theta}).$$

The maximum likelihood estimator $\hat{\boldsymbol{\theta}}_n$, which satisfies $\Psi_n(\hat{\boldsymbol{\theta}}_n) = \mathbf{0}$, is therefore an example of an M-estimator. See section (1.3) for additional examples. ♠

1.2 Asymptotics

Theorem 1.1 (Consistency). Suppose $(\mathbf{y}_i)_{i=1}^n$ is a random sample from F , and that $\boldsymbol{\theta}$ belongs to a compact parameter space Θ . Assume that:

- i. $\Psi_F(\boldsymbol{\theta})$ exists for all $\boldsymbol{\theta} \in \Theta$, and that $\boldsymbol{\theta}_0$ is the unique zero of $\Psi_F(\boldsymbol{\theta})$.
- ii. Each component of $\boldsymbol{\psi}(\mathbf{y}; \boldsymbol{\theta})$ is continuous in $\boldsymbol{\theta}$ and bounded by an integrable function of $\mathbf{y}$, not depending on $\boldsymbol{\theta}$.

If $\Psi_n(\hat{\boldsymbol{\theta}}_n) \xrightarrow{as} \mathbf{0}$, then $\hat{\boldsymbol{\theta}}_n \xrightarrow{as} \boldsymbol{\theta}_0$. □

Remark 1.1. See Boos and Stefanski (2013), theorem 7.1. ◆

Theorem 1.2 (Asymptotic Normality). Suppose $(\mathbf{y}_i)_{i=1}^n$ is a random sample from F , and $\boldsymbol{\theta}$ belongs to a compact parameter space Θ . Assume that:

- i. $\Psi_F(\boldsymbol{\theta})$ exists for all $\boldsymbol{\theta} \in \Theta$, and that $\boldsymbol{\theta}_0$ is the unique zero of $\Psi_F(\boldsymbol{\theta})$.
- ii. $\boldsymbol{\psi}(\mathbf{y}; \boldsymbol{\theta})$ is continuous and twice differentiable with respect to $\boldsymbol{\theta}$ for all $\mathbf{y}$ in the support of F and $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}_0$.
- iii. For $\boldsymbol{\theta}$ in a neighborhood of $\boldsymbol{\theta}_0$, there exists an integrable function $g(\mathbf{y})$ such that:

$$\left| \frac{\partial^2}{\partial \theta_{j_1} \partial \theta_{j_2}} \psi_{j_3}(\mathbf{y}; \boldsymbol{\theta}) \right| \leq g(\mathbf{y})$$

for $\forall (j_1, j_2, j_3) \in \{1, \dots, \dim(\boldsymbol{\theta})\}^3$, and $\int g(\mathbf{y}) dF(\mathbf{y}) < \infty$.

- iv. $\mathbf{A}(\boldsymbol{\theta}_0)$ exists and is non-singular, where:

$$\mathbf{A}(\boldsymbol{\theta}_0) = -\mathbb{E}_F\{\dot{\boldsymbol{\psi}}(\mathbf{y}; \boldsymbol{\theta}_0)\} = -\int \dot{\boldsymbol{\psi}}(\mathbf{y}; \boldsymbol{\theta}_0) dF(\mathbf{y}).$$

- v. $\mathbf{B}(\boldsymbol{\theta}_0)$ exists and is finite, where:

$$\mathbf{B}(\boldsymbol{\theta}_0) = \mathbb{E}_F\{\boldsymbol{\psi}(\mathbf{y}; \boldsymbol{\theta}_0) \otimes \boldsymbol{\psi}(\mathbf{y}; \boldsymbol{\theta}_0)\} = \int \boldsymbol{\psi}(\mathbf{y}; \boldsymbol{\theta}_0) \otimes \boldsymbol{\psi}(\mathbf{y}; \boldsymbol{\theta}_0) dF(\mathbf{y}).$$

If $\Psi_n(\hat{\boldsymbol{\theta}}_n) = o_p(n^{-1/2})$, then:

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N\{\mathbf{0}, \mathbf{\Omega}(\boldsymbol{\theta}_0)\},$$

$$\mathbf{\Omega}(\boldsymbol{\theta}_0) = \mathbf{A}^{-1}(\boldsymbol{\theta}_0) \mathbf{B}(\boldsymbol{\theta}_0) \mathbf{A}^{-T}(\boldsymbol{\theta}_0). \quad \square$$

Remark 1.2. See Boos and Stefanski (2013), theorem 7.2. ◆

Proof. By Taylor expansion:

$$o_p(n^{-1/2}) = \Psi_n(\hat{\theta}_n) = \Psi_n(\theta_0) + \dot{\Psi}_n(\theta_0)(\hat{\theta}_n - \theta_0) + \mathbf{R}_n(\hat{\theta}_n - \theta_0),$$

where the remainder:

$$\mathbf{R}_n = \sum_{j=1}^{\dim(\theta)} \frac{\partial \dot{\Psi}_n}{\partial \theta_j} \Big|_{\theta=\theta_n^*} (\hat{\theta}_{n,j} - \theta_{0,j})$$

and $\|\theta_n^* - \theta_0\| \leq \|\hat{\theta}_n - \theta_0\|$. By condition (iii.), the matrix of partial derivatives appearing in the remainder is bounded by a function not depending on θ , hence:

$$\|\mathbf{R}_n\| = \mathcal{O}_p(\|\hat{\theta}_n - \theta_0\|).$$

Since the conditions for consistency are contained within the conditions for asymptotic normality, $\hat{\theta}_n - \theta_0 = o_p(1)$, which implies $\mathbf{R}_n = o_p(1)$. Now:

$$\begin{aligned} o_p(n^{-1/2}) &= \Psi_n(\theta_0) + \{\dot{\Psi}_n(\theta_0) + o_p(1)\}(\hat{\theta}_n - \theta_0), \\ \sqrt{n}(\hat{\theta}_n - \theta_0) &= \{-\dot{\Psi}_n(\theta_0) + o_p(1)\}^{-1} \sqrt{n}\Psi_n(\theta_0). \end{aligned}$$

By the LLN:

$$-\dot{\Psi}_n(\theta_0) + o_p(1) \xrightarrow{p} -\mathbb{E}\{\dot{\psi}(\mathbf{y}; \theta_0)\} = \mathbf{A}(\theta_0)$$

By continuous mapping:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = \sqrt{n}\mathbf{A}^{-1}(\theta_0)\Psi_n(\theta_0) + o_p(1) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{\mathbf{A}^{-1}(\theta_0)\psi(\mathbf{y}_i; \theta_0)\} + o_p(1).$$

Identify $\varphi(\mathbf{y}_i; \theta_0) = \mathbf{A}^{-1}(\theta_0)\psi(\mathbf{y}_i; \theta_0)$ as the *influence function* of $\hat{\theta}_n$. By the CLT,

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{d} N\{\mathbf{0}, \Omega(\theta_0)\},$$

where:

$$\Omega(\theta_0) = \mathbb{V}\{\varphi(\mathbf{y}_i; \theta_0)\} = \mathbf{A}^{-1}(\theta_0)\mathbb{E}\{\psi(\mathbf{y}_i; \theta_0) \otimes \psi(\mathbf{y}_i; \theta_0)\}\mathbf{A}^{-T}(\theta_0).$$

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1.3 Examples

Example 1.1 (Ordinary Least Squares). Consider the model:

$$Y_i = \mathbf{x}_i' \beta + \epsilon_i,$$

where $(\epsilon_i)_{i=1}^n$ are independent, random residuals with $\mathbb{E}(\epsilon_i|\mathbf{x}_i) = 0$ and $\mathbb{V}(\epsilon_i|\mathbf{x}_i) = \sigma^2$. Define the squared error objective function:

$$Q(\boldsymbol{\beta}) = \frac{1}{2n} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \boldsymbol{\beta})^2.$$

Differentiating to obtain the estimating equation:

$$\boldsymbol{\Psi}_n(\boldsymbol{\beta}) = -\frac{\partial Q(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -\frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \boldsymbol{\beta}) \mathbf{x}_i.$$

Identify $\boldsymbol{\psi}(Y_i, \mathbf{x}_i; \boldsymbol{\beta}) = (Y_i - \mathbf{x}_i' \boldsymbol{\beta}) \mathbf{x}_i$. Let $\hat{\boldsymbol{\beta}}$ denote a solution to $\boldsymbol{\Psi}_n(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, then under the assumptions of theorem (1.2), $\hat{\boldsymbol{\beta}}$ is an M -estimator.

The partial of $\boldsymbol{\psi}$ with respect to $\boldsymbol{\beta}'$ is:

$$\frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\beta}'} = -\mathbf{x}_i \otimes \mathbf{x}_i.$$

$\mathbf{A}(\boldsymbol{\theta}_0)$ takes the form:

$$\mathbf{A}(\boldsymbol{\theta}_0) \equiv -\mathbb{E}_0 \left(\frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\beta}'} \right) = \mathbb{E}_0(\mathbf{x}_i \otimes \mathbf{x}_i).$$

The empirical estimate of $\mathbf{A}(\boldsymbol{\theta}_0)$ is:

$$\hat{\mathbf{A}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \otimes \mathbf{x}_i.$$

The outer product of $\boldsymbol{\psi}$ is:

$$\boldsymbol{\psi} \otimes \boldsymbol{\psi} = (Y_i - \mathbf{x}_i' \boldsymbol{\beta})^2 \mathbf{x}_i \otimes \mathbf{x}_i.$$

$\mathbf{B}(\boldsymbol{\beta}_0)$ takes the form:

$$\mathbf{B}(\boldsymbol{\beta}_0) = \mathbb{E}_0(\boldsymbol{\psi} \otimes \boldsymbol{\psi}) = \mathbb{E}_0\{(Y_i - \mathbf{x}_i' \boldsymbol{\beta})^2 \mathbf{x}_i \otimes \mathbf{x}_i\} = \sigma^2 \mathbb{E}_0(\mathbf{x}_i \otimes \mathbf{x}_i).$$

The empirical estimate of $\mathbf{B}(\boldsymbol{\beta}_0)$ is:

$$\hat{\mathbf{B}} = \hat{\sigma}^2 \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \otimes \mathbf{x}_i = \hat{\sigma}^2 \mathbf{X}' \mathbf{X},$$

where $\hat{\sigma}^2$ is any consistent estimate of the residual variance.

The asymptotic approximation to the distribution of $\hat{\boldsymbol{\beta}}$ is:

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}_0, n^{-1} \hat{\boldsymbol{\Omega}}),$$

where $\hat{\boldsymbol{\Omega}} = \hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-T}$.



Example 1.2 (Non-linear Least Squares). Consider the model:

$$Y_i = g(\mathbf{x}_i, \boldsymbol{\beta}) + \epsilon_i,$$

where g is a known differentiable function, and $(\epsilon_i)_{i=1}^n$ are independent, random residuals with $\mathbb{E}(\epsilon_i|\mathbf{x}_i) = 0$ and $\mathbb{V}(\epsilon_i|\mathbf{x}_i) = \sigma_i^2$. Define the squared error objective function:

$$Q(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \{Y_i - g(\mathbf{x}_i, \boldsymbol{\beta})\}^2.$$

Differentiating to obtain the estimating equation:

$$\boldsymbol{\Psi}_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n \{Y_i - g(\mathbf{x}_i, \boldsymbol{\beta})\} \dot{g}(\mathbf{x}_i, \boldsymbol{\beta}).$$

Identify $\boldsymbol{\psi}(Y_i, \mathbf{x}_i; \boldsymbol{\beta}) = \{Y_i - g(\mathbf{x}_i, \boldsymbol{\beta})\} \dot{g}(\mathbf{x}_i, \boldsymbol{\beta})$. Let $\hat{\boldsymbol{\beta}}$ denote a solution to $\boldsymbol{\Psi}_n(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, then under the assumptions of theorem (1.2), $\hat{\boldsymbol{\beta}}$ is an M -estimator.

$\mathbf{A}(\boldsymbol{\beta}_0)$ takes the form:

$$\mathbf{A}(\boldsymbol{\beta}_0) = -\mathbb{E}_0\{\dot{\boldsymbol{\psi}}(Y_i, \mathbf{x}_i; \boldsymbol{\beta}_0)\} = \mathbb{E}_0\{\dot{g}(\mathbf{x}_i, \boldsymbol{\beta}_0) \otimes \dot{g}(\mathbf{x}_i, \boldsymbol{\beta}_0)\}.$$

The empirical estimate of $\mathbf{A}(\boldsymbol{\beta}_0)$ is:

$$\hat{\mathbf{A}} = \frac{1}{n} \sum_{i=1}^n \dot{g}(\mathbf{x}_i, \hat{\boldsymbol{\beta}}) \otimes \dot{g}(\mathbf{x}_i, \hat{\boldsymbol{\beta}}).$$

$\mathbf{B}(\boldsymbol{\beta}_0)$ takes the form:

$$\mathbf{B}(\boldsymbol{\beta}_0) = \mathbb{E}_0\left[\{Y_i - g(\mathbf{x}_i, \boldsymbol{\beta}_0)\}^2 \dot{g}(\mathbf{x}_i, \boldsymbol{\beta}_0) \otimes \dot{g}(\mathbf{x}_i, \boldsymbol{\beta}_0)\right] = \sigma_i^2 \mathbb{E}_0\{\dot{g}(\mathbf{x}_i, \boldsymbol{\beta}_0) \otimes \dot{g}(\mathbf{x}_i, \boldsymbol{\beta}_0)\}.$$

The empirical estimate of $\mathbf{B}(\boldsymbol{\beta}_0)$ is:

$$\hat{\mathbf{B}} = \frac{1}{n} \sum_{i=1}^n \{Y_i - g(\mathbf{x}_i, \hat{\boldsymbol{\beta}})\}^2 \dot{g}(\mathbf{x}_i, \hat{\boldsymbol{\beta}}) \otimes \dot{g}(\mathbf{x}_i, \hat{\boldsymbol{\beta}}).$$

The asymptotic approximation to the distribution of $\hat{\boldsymbol{\beta}}$ is:

$$\hat{\boldsymbol{\beta}} \dot{\sim} N(\boldsymbol{\beta}_0, n^{-1}\hat{\boldsymbol{\Omega}}),$$

where $\hat{\boldsymbol{\Omega}} = \hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-T}$. ♠

Example 1.3 (Robust Regression). Consider again the model:

$$Y_i = \mathbf{x}_i' \boldsymbol{\beta} + \epsilon_i,$$

where $(\epsilon_i)_{i=1}^n$ are independent, random residuals with $\mathbb{E}(\epsilon_i|\mathbf{x}_i) = 0$ and $\mathbb{V}(\epsilon_i|\mathbf{x}_i) = \sigma_i^2$.

The ordinary least squares estimating equations are:

$$\Psi_n(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \boldsymbol{\beta}) \mathbf{x}_i \stackrel{\text{Set}}{=} \mathbf{0}.$$

A robust, alternative set of estimating equations is:

$$\Psi_n(\boldsymbol{\beta}) = \sum_{i=1}^n \psi_\tau(Y_i - \mathbf{x}_i' \boldsymbol{\beta}) \mathbf{x}_i.$$

Here ψ_τ is a bounded score function, such as Huber's function:

$$\psi_\tau(t) = \begin{cases} -\tau, & t < -\tau, \\ t, & -\tau \leq t \leq \tau, \\ \tau, & t > \tau. \end{cases}$$

or Tukey's biweight function:

$$\psi_\tau(t) = \begin{cases} t \left(1 - \frac{t^2}{\tau^2}\right)^2, & |t| < \tau, \\ 0, & |t| > \tau. \end{cases}$$

Let $\hat{\boldsymbol{\beta}}$ denote a solution to $\Psi_n(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, then under the assumptions of theorem (1.2), $\hat{\boldsymbol{\beta}}$ is an M -estimator.

$\mathbf{A}(\boldsymbol{\beta}_0)$ takes the form:

$$\mathbf{A}(\boldsymbol{\beta}_0) = -\mathbb{E}_0\{\dot{\psi}_\tau(Y_i - \mathbf{x}_i' \boldsymbol{\beta}_0) \mathbf{x}_i \otimes \mathbf{x}_i\}.$$

The empirical estimate of $\mathbf{A}(\boldsymbol{\beta}_0)$ is:

$$\hat{\mathbf{A}} = \frac{1}{n} \sum_{i=1}^n \dot{\psi}_\tau(Y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}) \mathbf{x}_i \otimes \mathbf{x}_i.$$

$\mathbf{B}(\boldsymbol{\beta}_0)$ takes the form:

$$\mathbf{B}(\boldsymbol{\beta}_0) = \mathbb{E}_0\{\psi_\tau^2(Y_i - \mathbf{x}_i' \boldsymbol{\beta}_0) \mathbf{x}_i \otimes \mathbf{x}_i\}.$$

The empirical estimate of $\mathbf{B}(\boldsymbol{\beta}_0)$ is:

$$\hat{\mathbf{B}} = \frac{1}{n} \sum_{i=1}^n \psi_\tau^2(Y_i - \mathbf{x}_i' \hat{\boldsymbol{\beta}}) \mathbf{x}_i \otimes \mathbf{x}_i.$$

The asymptotic approximation to the distribution of $\hat{\boldsymbol{\beta}}$ is:

$$\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}_0, n^{-1} \hat{\boldsymbol{\Omega}}),$$

where $\hat{\boldsymbol{\Omega}} = \hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-T}$.

